STAT 437
Midterm project

Group 5

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Introduction

Introduction

Purpose of the study: to identify influence factors associated with coronary heart disease.

Total observation: 609 collected from white males in Evans Country who were followed for 7 years.

Outcome of interest: Coronary heart disease

Data source: data set “Evans” which available in r-package “lbreg”.

Variables (9 variables in total): Country heart disease(CDH), catecholamine level(CAT), age(AGE), cholesterol(CHL), subject has ever smoked(SMK), electrocardiogram abnormality(EGG), diastolic blood pressure(DBP), systolic blood pressure(SBP), SBP greater than or equal to 160 or DBP greater or equal to 95 (HPT).

Design of choice: Binary logistic regression

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Method for Hypothesis

Method for hypothesis — compare two nested models

Compare two nested models

H_o : there is no difference between the two models

H_1 : there is a difference between the models

$$D_R - D_F \rightarrow \chi^2(\mathrm{d}), \mathrm{n} \rightarrow \infty$$

D_R : deviance for the restricted model

D_F : deviance for the full model

Method for hypothesis — find over-dispersion

Method for finding over-dispersion

For a logistic regression model:

$$\mathrm{Var}(y_i) = \tau \mathrm{E}(y_i)(1 - \mathrm{E}(y_i))$$

τ:dispersion parameter

For τ=1, y_i followers a binomial distribution

For τ≠1, y_i does not follow a binomial distribution

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Data Analysis

Data Analysis — FItting the model

```
log = glm(CDH ~ ., family = binomial, data = evans)
summary(log)
```

```
>
> Call:
> glm(formula = CDH ~ ., family = binomial, data = evans)
>
> Deviance Residuals:
>     Min       1Q   Median       3Q      Max
> -1.1291  -0.5222  -0.4033  -0.2835   2.6444
>
> Coefficients:
>              Estimate Std. Error z value Pr(>|z|)
> (Intercept) -6.764709   1.579763  -4.282 1.85e-05 ***
> CATTRUE      0.750257   0.377231   1.989  0.04672 *
> AGE          0.037781   0.016031   2.357  0.01844 *
> CHL          0.008674   0.003296   2.632  0.00850 **
> SMKTRUE      0.867474   0.307218   2.824  0.00475 **
> ECGTRUE      0.403426   0.296125   1.362  0.17309
> DBP          0.012402   0.014284   0.868  0.38526
> SBP         -0.010571   0.008073  -1.309  0.19038
> HPTTRUE      0.532752   0.383747   1.388  0.16505
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> (Dispersion parameter for binomial family taken to be 1)
>
>     Null deviance: 438.56  on 608  degrees of freedom
> Residual deviance: 398.60  on 600  degrees of freedom
> AIC: 416.6
>
> Number of Fisher Scoring iterations: 5
```

Logistic Regression model behaves better when it comes to modeling and predicting binary or categorical outcomes compared to linear regression model.

Data Analysis — Over-dispersion

```
DIS = sum(residuals(log, type="pearson")^2)/log$df.res
print(DIS)
```

```
> [1] 1.002793
```

DIS = 1.002793 roughly equal to 1

↓

No over-dispersion problem exists

↓

The result from model log may be reliable.

Data analysis — Variable Selection Using Stepwise AIC values

```
log2 = stepAIC(log, direction = c("both"), k = 2)

summary(log2)
```

```
>
> Call:
> glm(formula = CDH ~ CAT + AGE + CHL + SMK + HPT, family = binomial,
>     data = evans)
>
> Deviance Residuals:
>     Min       1Q   Median       3Q      Max
> -1.0762  -0.5297  -0.3995  -0.3048   2.5356
>
> Coefficients:
>             Estimate Std. Error z value Pr(>|z|)
> (Intercept) -6.680112   1.136363  -5.879 4.14e-09 ***
> CATTRUE      0.715810   0.340180   2.104  0.03536 *
> AGE          0.032770   0.015197   2.156  0.03105 *
> CHL          0.008608   0.003259   2.641  0.00827 **
> SMKTRUE      0.802906   0.303001   2.650  0.00805 **
> HPTTRUE      0.476272   0.289296   1.646  0.09970 .
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> (Dispersion parameter for binomial family taken to be 1)
>
>     Null deviance: 438.56  on 608  degrees of freedom
> Residual deviance: 401.95  on 603  degrees of freedom
> AIC: 413.95
>
> Number of Fisher Scoring iterations: 5
```

Based on stepwise AIC value, EGG, DBP, SBP are removed.

Data Analysis — checking for Covariate Significance(1)

```
anova(log2, test="Chi")
```

```
> Analysis of Deviance Table
>
> Model: binomial, link: logit
>
> Response: CDH
>
> Terms added sequentially (first to last)
>
>
>      Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
> NULL                   608     438.56
> CAT   1  14.1312       607     424.43 0.0001705 ***
> AGE   1   4.0483       606     420.38 0.0442171 *
> CHL   1   8.2166       605     412.16 0.0041510 **
> SMK   1   7.5056       604     404.66 0.0061506 **
> HPT   1   2.7097       603     401.95 0.0997410 .
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
'.' 0.1 ' ' 1
```

For checking for Covariate Significance, Wald-type is not used due to presence of categorical variables. Comparing Step-wise Nested Models using Chi-Square Hypothesis Tests is used.

All covariates are significant in log2 at a 0.1 significance level. (HPT not significant at 0.05 level)

Data Analysis — checking for Covariate Significance(2)

New model without HPT:

```
log3 = glm(CDH~CAT+ AGE+ CHL+ SMK, family = binomial, data = evans)

summary(log3)
```

```
>
> Call:
> glm(formula = CDH ~ CAT + AGE + CHL + SMK, family = binomial,
>     data = evans)
>
> Deviance Residuals:
>     Min       1Q   Median       3Q      Max
> -1.0562  -0.5230  -0.4064  -0.3150   2.4855
>
> Coefficients:
>             Estimate Std. Error z value Pr(>|z|)
> (Intercept) -6.659008   1.127218  -5.907 3.47e-09 ***
> CATTRUE      0.931183   0.315154   2.955  0.00313 **
> AGE          0.033141   0.015171   2.184  0.02893 *
> CHL          0.009294   0.003228   2.879  0.00398 **
> SMKTRUE      0.789630   0.301677   2.617  0.00886 **
> ---
> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> (Dispersion parameter for binomial family taken to be 1)
>
>     Null deviance: 438.56  on 608  degrees of freedom
> Residual deviance: 404.66  on 604  degrees of freedom
> AIC: 414.66
>
> Number of Fisher Scoring iterations: 5
```

Data Analysis — Model Selection by comparing the Nested Models

H_0: There are no differences between the models log2 and log.

H_A: There are some differences between the models log2 and log.

```
anova(log2, log)
```

```
> Analysis of Deviance Table
>
> Model 1: CDH ~ CAT + AGE + CHL + SMK + HPT
> Model 2: CDH ~ CAT + AGE + CHL + SMK + ECG + DBP + SBP + HPT
>   Resid. Df Resid. Dev Df Deviance
> 1       603     401.95
> 2       600     398.60  3   3.3468
```

Difference in deviance = 3.3468

Degrees of freedom = 3

$P(\chi^2_3 > 3.3468) = 0.34117$

P-value > $\alpha = 0.05$

At 5% significance level, do not reject H_0

Data Analysis — Model Selection by Comparing the Restricted Nested Models

H_0: There are no differences between the models log3 and log2.

H_A: There are some differences between the models log3 and log2.

```
anova(log3, log2)
```

```
> Analysis of Deviance Table
>
> Model 1: CDH ~ CAT + AGE + CHL + SMK
> Model 2: CDH ~ CAT + AGE + CHL + SMK + HPT
>   Resid. Df Resid. Dev Df Deviance
> 1       604     404.66
> 2       603     401.95  1   2.7097
```

Difference in deviances = 2.7097

Degree of freedom = 1

$P(\chi^2_1 > 2.7097) = 0.09977$

P-value > $\alpha = 0.05$

At 5% significance levle, do not reject H_0

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Conclusion

Conclusion

(a) By fitting binary logistic regression as a full model, no over-dispersion problem is found.

(b) Using stepAIC() command, we identify the most important predictors for predicting the response variables are CAT, AGE, CHL, SMK, HPT.

(c) Significant covariates are CAT, AGE, CHL, and SMK.

(d) By comparing one full model and one restricted model, the model selection shows the restricted model log3 is the best fitted.

Reference

http://web1.sph.emory.edu/dkleinb/logreg3.htm#data